

```

<?xml version="1.0" encoding="UTF-8"?>
<svg viewBox="0 0 800 600" xmlns="http://www.w3.org/2000/svg">
  <!-- Background -->
  <rect width="800" height="600" fill="#f8f9fa" />

  <!-- Title -->
  <text x="400" y="40" font-family="Arial, sans-serif" font-size="24" font-weight
    PinkSync System Architecture
  </text>
  <text x="400" y="65" font-family="Arial, sans-serif" font-size="16" text-anchor
    Bidirectional Sign Language Authentication & Communication System
  </text>

  <!-- Main System Box -->
  <rect x="100" y="100" width="600" height="450" rx="15" fill="none" stroke="#333

  <!-- System Components -->

  <!-- Input Capture Module -->
  <rect x="125" y="125" width="160" height="100" rx="10" fill="#ffcad4" stroke="#"
  <text x="205" y="145" font-family="Arial, sans-serif" font-size="14" font-weigh
    Input Capture
  </text>
  <text x="205" y="165" font-family="Arial, sans-serif" font-size="12" text-anch
    • 3D Optical Sensors
  </text>
  <text x="205" y="185" font-family="Arial, sans-serif" font-size="12" text-anch
    • 43-Point Facial Tracking
  </text>
  <text x="205" y="205" font-family="Arial, sans-serif" font-size="12" text-anch
    • Spatial Movement Analysis
  </text>

  <!-- Authentication Module -->
  <rect x="320" y="125" width="160" height="100" rx="10" fill="#ffc6ff" stroke="#"
  <text x="400" y="145" font-family="Arial, sans-serif" font-size="14" font-weigh
    Authentication Module
  </text>
  <text x="400" y="165" font-family="Arial, sans-serif" font-size="12" text-anch
    • Sign Biometric Verification
  </text>
  <text x="400" y="185" font-family="Arial, sans-serif" font-size="12" text-anch
    • Motion-Based Passwords
  </text>

```

```

<text x="400" y="205" font-family="Arial, sans-serif" font-size="12" text-anch
  • Visual 2FA
</text>

<!-- Cultural Integration Layer -->
<rect x="515" y="125" width="160" height="100" rx="10" fill="#bde0fe" stroke="#"
<text x="595" y="145" font-family="Arial, sans-serif" font-size="14" font-weigh
  Cultural Integration
</text>
<text x="595" y="165" font-family="Arial, sans-serif" font-size="12" text-anch
  • Regional Sign Variants
</text>
<text x="595" y="185" font-family="Arial, sans-serif" font-size="12" text-anch
  • Community Terminology
</text>
<text x="595" y="205" font-family="Arial, sans-serif" font-size="12" text-anch
  • Deaf Epistemology
</text>

<!-- Processing Engine -->
<rect x="125" y="250" width="550" height="120" rx="10" fill="#a3c4f3" stroke="#"
<text x="400" y="270" font-family="Arial, sans-serif" font-size="14" font-weigh
  Spatial Grammar Processing Engine
</text>

<!-- Processing Components -->
<rect x="150" y="290" width="150" height="65" rx="5" fill="#b8c0ff" stroke="#43
<text x="225" y="310" font-family="Arial, sans-serif" font-size="12" font-weigh
  3D Coordinate Mapping
</text>
<text x="225" y="330" font-family="Arial, sans-serif" font-size="10" text-anch
  Maps signing space usage
</text>

<rect x="325" y="290" width="150" height="65" rx="5" fill="#b8c0ff" stroke="#43
<text x="400" y="310" font-family="Arial, sans-serif" font-size="12" font-weigh
  Non-Manual Recognition
</text>
<text x="400" y="330" font-family="Arial, sans-serif" font-size="10" text-anch
  Facial grammar markers
</text>

<rect x="500" y="290" width="150" height="65" rx="5" fill="#b8c0ff" stroke="#43
<text x="575" y="310" font-family="Arial, sans-serif" font-size="12" font-weigh
  Classifier Prediction
</text>
<text x="575" y="330" font-family="Arial, sans-serif" font-size="10" text-anch

```

Visual descriptive elements

</text>

<!-- Translation Module -->

<rect x="125" y="395" width="250" height="130" rx="10" fill="#caffff" stroke="#

<text x="250" y="415" font-family="Arial, sans-serif" font-size="14" font-weigh

Bidirectional Translation

</text>

<!-- Arrows for bidirectional -->

<line x1="190" y1="435" x2="190" y2="485" stroke="#38b000" stroke-width="2" />

<polygon points="185,435 190,425 195,435" fill="#38b000" />

<polygon points="185,485 190,495 195,485" fill="#38b000" />

<text x="240" y="455" font-family="Arial, sans-serif" font-size="12" text-anch

Sign Language

</text>

<text x="240" y="475" font-family="Arial, sans-serif" font-size="12" text-anch

↔

</text>

<text x="240" y="495" font-family="Arial, sans-serif" font-size="12" text-anch

Spoken/Written Language

</text>

<!-- Visual Information Architecture -->

<rect x="425" y="395" width="250" height="130" rx="10" fill="#fdfdb6" stroke="#

<text x="550" y="415" font-family="Arial, sans-serif" font-size="14" font-weigh

Visual-Spatial Architecture

</text>

<text x="550" y="440" font-family="Arial, sans-serif" font-size="12" text-anch

• Spatial Organization Interface

</text>

<text x="550" y="460" font-family="Arial, sans-serif" font-size="12" text-anch

• Visual Memory Mapping

</text>

<text x="550" y="480" font-family="Arial, sans-serif" font-size="12" text-anch

• Visual Search Engine

</text>

<text x="550" y="500" font-family="Arial, sans-serif" font-size="12" text-anch

• Cognitive Load Optimization

</text>

<!-- Devices -->

<circle cx="180" y="570" r="15" fill="#d8d8d8" stroke="#333" stroke-width="1" /

<rect x="165" y="590" width="30" height="3" rx="1" fill="#333" />

<text x="210" y="580" font-family="Arial, sans-serif" font-size="12" text-anch

Mobile

</text>

```

<rect x="290" y="560" width="30" height="20" rx="2" fill="#d8d8d8" stroke="#333
<rect x="280" y="580" width="50" height="3" rx="1" fill="#333" />
<text x="340" y="580" font-family="Arial, sans-serif" font-size="12" text-anch
    Desktop
</text>

<rect x="430" y="563" width="25" height="15" rx="2" fill="#d8d8d8" stroke="#333
<rect x="425" y="578" width="35" height="10" rx="1" fill="#d8d8d8" stroke="#333
<text x="470" y="580" font-family="Arial, sans-serif" font-size="12" text-anch
    Wearable
</text>

<rect x="570" y="560" width="25" height="18" rx="2" fill="#d8d8d8" stroke="#333
<rect x="562" y="578" width="40" height="5" rx="1" fill="#d8d8d8" stroke="#333"
<text x="610" y="580" font-family="Arial, sans-serif" font-size="12" text-anch
    IoT
</text>

<!-- Connection lines to devices -->
<line x1="250" y1="550" x2="400" y2="550" stroke="#333" stroke-width="1.5" stro
<line x1="550" y1="550" x2="400" y2="550" stroke="#333" stroke-width="1.5" stro
<line x1="180" y1="550" x2="250" y2="550" stroke="#333" stroke-width="1" />
<line x1="305" y1="550" x2="305" y2="560" stroke="#333" stroke-width="1" />
<line x1="442" y1="550" x2="442" y2="563" stroke="#333" stroke-width="1" />
<line x1="582" y1="550" x2="582" y2="560" stroke="#333" stroke-width="1" />
<line x1="180" y1="550" x2="180" y2="555" stroke="#333" stroke-width="1" />

<!-- Key Points vs Existing Patent -->
<rect x="15" y="40" width="180" height="40" rx="5" fill="#ff9e80" stroke="#dd2c
<text x="105" y="65" font-family="Arial, sans-serif" font-size="12" font-weight
    Unique vs. Existing Patent
</text>

<!-- Connection lines from unique points -->
<line x1="195" y1="60" x2="320" y2="125" stroke="#dd2c00" stroke-width="1.5" st
<line x1="195" y1="60" x2="515" y2="125" stroke="#dd2c00" stroke-width="1.5" st
<line x1="195" y1="60" x2="400" y2="250" stroke="#dd2c00" stroke-width="1.5" st
<line x1="195" y1="60" x2="550" y2="395" stroke="#dd2c00" stroke-width="1.5" st
</svg>

```